



**REVA**  
UNIVERSITY  
Bengaluru, India



## **School of Computer Science and Engineering**

### **MINI PROJECT REPORT ON**

### **FitLens AI: Efficient Body Measurements System**

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**Course Name: Mini Project-Research Based**  
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**In partial fulfillment of the requirements for the award of the degree of**

**BACHELOR OF TECHNOLOGY  
IN  
Information Science and Engineering**

**Under the guidance of**

**Dr.Argha Sarkar**  
Associate Professor

**Academic Year: 2025-26**



**REVA**  
UNIVERSITY  
Bengaluru, India

## **School of Computer Science and Engineering**

### **CERTIFICATE**

This is to certify that the mini project B22CI0605 entitled “FitLens AI: Efficient Body measurements system” is a bona fide work carried out by

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In partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Information Science Engineering of REVA University, Bengaluru, during the academic year 2025-2026.

This work has been carried out under my supervision and guidance. To the best of my knowledge, this work has not been submitted previously for the award of any degree or diploma.

#### **Signature of Guide**

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## DECLARATION

We hereby declare that the mini project entitled “FitLens AI: Efficient Body Measurements system” is a bona fide work carried out by us in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Information Science Engineering of REVA University, Bengaluru, during the academic year 2025-26.

We further declare that this work has not been submitted previously for the award of any degree or diploma in any university or institution. All sources of information used in this project have been properly acknowledged.

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## ABSTRACT

High return rates in the e-commerce apparel industry — often exceeding 30% — are largely driven by poor clothing fit and inaccurate size selection. Existing solutions, such as 3D scanners, depth cameras, or manual tape measurements, either require expensive hardware, are prone to human error, or fail to scale for online use. This project presents FitLens AI, a camera-vision-based body measurement system that extracts accurate anthropometric measurements from standard 2D photographs, requiring no specialized hardware beyond a regular smartphone or webcam.

FitLens AI employs a multi-stage computer vision pipeline that integrates YOLOv8n-seg for human detection and bounding box generation, MediaPipe Pose for 2D anatomical landmark extraction (33 keypoints), and a dynamic pixel-to-centimeter calibration module using user-provided height as a reference. The system supports both photo-upload and live camera modes, with real-time posture correction feedback delivered via color-coded skeleton overlays. A lightweight face verification mechanism ensures identity consistency across multiple captured frames.

The system was evaluated on 20 subjects (10 male, 10 female) against tape measurements taken by trained personnel following ISO 8559-1 standards. FitLens AI achieved a Mean Absolute Error (MAE) below 0.9 cm for linear measurements such as shoulder width and arm length, surpassing existing vision-based methods (which typically exhibit 4–8 cm errors). The system processes a full measurement in approximately 5 seconds on a standard CPU, operates entirely on-device with no cloud transmission, and is compatible with everyday consumer hardware. FitLens AI presents a practical, privacy-respecting, open-source alternative for virtual fitting rooms, online retail sizing, and remote fitness tracking.

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# CHAPTER 1: INTRODUCTION

## 1.1 Background of the Project

The global e-commerce apparel industry has grown exponentially over the past decade, with millions of customers purchasing clothing online without the ability to try it on physically. This has resulted in a persistent and costly challenge: an extremely high rate of product returns, often exceeding 30%, driven primarily by poor fit and incorrect size selection. The problem is not only a logistical and financial burden for retailers and consumers but also carries significant environmental implications — returned items contribute to increased carbon emissions from additional shipping, excess packaging waste, and the frequent disposal of merchandise that cannot be resold.

At the same time, the digital fitness industry and the growing adoption of telemedicine have created a new demand for tools that can measure the human body remotely and accurately. Fitness enthusiasts want to track body changes over time, and healthcare applications increasingly require body measurements for personalized health monitoring, all without the need for physical clinic visits.

Historically, body measurement has relied on manual anthropometry — trained individuals using tape measures to record dimensions such as chest, waist, and shoulder width. While reliable when performed correctly, this approach is inherently inconsistent, labor-intensive, and inaccessible to remote users. Technology-based alternatives, such as 3D body scanners, offer precision but cost between \$10,000 and \$150,000, placing them out of reach for the average consumer or small retailer.

Recent advances in computer vision, particularly in 2D pose estimation and keypoint detection using deep learning, have opened a new path. Frameworks such as MediaPipe and OpenPose now allow real-time extraction of body joint coordinates from standard RGB images on consumer hardware. FitLens AI is built upon this foundation, combining these technologies with a robust pixel-to-scale calibration system to deliver accurate body measurements using only a regular camera.

## 1.2 Problem Statement

There is currently no widely accessible, accurate, and privacy-preserving solution for obtaining body measurements in a remote or online context. Manual tape measurements are inconsistent and cannot be performed reliably by an untrained user alone. Commercial 3D scanning systems are prohibitively expensive. Existing vision-based software tools achieve limited accuracy (errors of 4–8 cm) and often rely on proprietary cloud infrastructure, raising data privacy concerns. The gap between the need for accurate body measurements and consumer-accessible tools remains unresolved.

## 1.3 Objectives

The core objectives of the FitLens AI project are:

- To design and implement a multi-stage computer vision pipeline capable of detecting a person in frame, extracting 2D anatomical landmarks, and computing body measurements from standard RGB images or live video.
- To develop a real-time posture correction feedback mechanism using color-coded skeleton overlays and plain-language instructions to guide users into a standardized measurement pose.
- To accurately compute key anthropometric measurements, including shoulder width, arm length, torso length, hip width, chest width, and leg length, using 2D landmark coordinates and dynamic pixel-to-scale mapping.
- To achieve a geometric error margin of 1–3 cm for all measurements, significantly outperforming existing vision-based methods that exhibit errors of  $\pm 4$ –8 cm.
- To implement a privacy-first architecture that performs all computation on-device with no biometric data transmitted to or stored in the cloud.

## **1.4 Scope of the Project**

FitLens AI is designed to operate as a web-based application accessible from any standard browser on a laptop or smartphone. The system accepts two types of input: a pair of uploaded photographs (front and side views) or a live camera stream. It computes the following body measurements: height reference, shoulder width, arm length, torso length, hip width, and inseam/leg length. Circumferential measurements (chest, waist, hip) are estimated using an elliptical approximation from the front and side views.

The system is scoped for use by adult individuals aged 18–60 with heights between 150 cm and 200 cm, under standard indoor lighting conditions. The camera should be positioned approximately 1.5–2 meters from the subject. The system does not support measurement of children, seated individuals, or subjects with significant body occlusion from furniture or other objects.

The project does not include integration with any specific e-commerce platform or retail backend, though the JSON output format is designed to be extensible for such integrations.

## **1.5 Organization of the Report**

This report is organized as follows. Chapter 2 provides a comprehensive literature survey covering existing body measurement methods, their limitations, and the state of the art in computer vision-based approaches. Chapter 3 presents a system analysis comparing the existing system with the proposed FitLens AI solution. Chapter 4 details the system design, including architecture diagrams, data flow diagrams, and key algorithms. Chapter 5 describes the implementation, covering tools, technologies, hardware requirements, and module-level descriptions. Chapter 6 presents the results and performance evaluation. Chapter 7 concludes the report with key findings, identified limitations, and directions for future work.

## **CHAPTER 2: LITERATURE SURVEY**

This chapter reviews the existing approaches to body measurement, their methodologies, and their limitations. Understanding prior work is essential to situate FitLens AI within the broader research landscape and to justify the design decisions made in this project.

### **2.1 Traditional and Manual Measurement Methods**

The most widely used method for body measurement remains the manual tape measure. In the tailoring and apparel industry, this technique has been standardized through bodies such as ASTM International and ISO (ISO 8559), which specify the landmark points on the body and the required posture during measurement. When performed by a trained professional, tape measurements can achieve reasonable consistency. However, the approach has several well-documented limitations.

Human error is a significant factor. The final measurement depends on the operator's technique — tape tension, identification of the correct anatomical landmark, and reading accuracy all introduce variability. Studies have shown that inter-observer variability in retail settings can reach up to 5 cm for a single measurement. Fatigue further compounds this issue when many subjects are measured in succession.

Self-measurement, which is what most online shoppers are expected to do, is even less reliable. It is physically difficult to apply a tape measure correctly to one's own body without assistance. Positioning errors, improper tension, and incorrect landmark identification are common, resulting in measurements that may differ significantly from those taken by a professional.

Finally, the physical presence of the user is mandatory — a fundamental incompatibility with the remote, online context in which body measurements are increasingly needed.

### **2.2 Computer Vision and Depth-Based Measurement Systems**

Recognizing the limitations of manual methods, researchers and product developers have explored technology-based alternatives. Early approaches centered on 3D body scanners, which use arrays of structured light, laser triangulation, or stereo cameras to create a full 3D model of the body. Systems such as the [TC]<sup>2</sup> NX-16 can deliver accuracy within 1–2 cm across all body dimensions. However, the hardware cost (ranging from \$10,000 to \$150,000), the requirement

for a controlled scanning booth, and the need for trained operators make these systems inaccessible to most consumers and small retailers.

Consumer-grade depth cameras, such as the Microsoft Kinect and Intel RealSense, offered a lower-cost entry point into 3D body measurement. These devices can capture depth maps of the scene in real time, enabling body volume estimation without a dedicated scanning booth. However, depth cameras still require specialized hardware, have limited range and accuracy in typical indoor lighting, and are sensitive to clothing texture, which can distort depth readings.

The emergence of 2D pose estimation frameworks marked a significant shift. OpenPose, developed by Carnegie Mellon University, demonstrated that real-time multi-person keypoint detection (covering 18–25 body joints) could be achieved on standard hardware using Convolutional Neural Networks (CNNs). MediaPipe Pose, developed by Google, extended this to a lightweight, cross-platform library capable of detecting 33 anatomical landmarks from a single RGB video frame at over 30 frames per second on a mobile device. These frameworks made body keypoint data accessible to any developer with a standard camera, opening the door to software-only body measurement solutions.

YOLOv8, the latest iteration of the You Only Look Once object detection family developed by Ultralytics, further complements pose estimation by providing fast and accurate person detection and segmentation. YOLOv8n-seg, the nano segmentation variant, is particularly suited to constrained hardware environments due to its small model size and high inference speed.

## **2.3 Comparison of Methods and Identified Limitations**

While 2D pose estimation frameworks dramatically improved accessibility, they introduced a new set of limitations specific to the monocular (single-camera) measurement context.

First, skeletal joint detection identifies the internal coordinates of body joints, but does not capture the external body silhouette. For measurements such as chest circumference or hip width, the system must estimate the body boundary, not just the joint position. Loose or oversized clothing further complicates this, as it obscures the true body contour.

Second, monocular images lack depth information. A 2D camera cannot directly measure the depth (front-to-back dimension) of the body, which is essential for circumferential

measurements. Heuristic or model-based approaches must be used to estimate these values from visible 2D data.

Third, pixel-to-real-world calibration is a persistent challenge. Without a known reference scale in the image (such as a calibration object or the user's height), converting pixel distances to centimeters is highly sensitive to camera focal length, distance, and perspective distortion.

Fourth, most existing vision-based systems require controlled conditions — uniform backgrounds, specific lighting, and a fixed camera distance — limiting their utility in real-world settings.

FitLens AI addresses all four limitations through its combination of user-height-based calibration, edge-contour refinement via YOLOv8n-seg, real-time posture validation, and multi-frame averaging for temporal stability.

## CHAPTER 3: SYSTEM ANALYSIS

### 3.1 Existing System

The existing approaches to body measurement for e-commerce and fitness applications can be grouped into three broad categories: manual tape measurements, commercial 3D scanning systems, and generic 2D pose estimation tools.

Manual tape measurements are the most widely used but suffer from high inter-observer variability (up to 5 cm), dependence on user skill, inability to scale online, and the requirement for physical presence. They are unsuitable for remote or self-service use.

Commercial 3D scanning systems (e.g., [TC]<sup>2</sup> NX-16, Styku) provide high accuracy (within 1–2 cm) but cost between \$10,000 and \$150,000, require dedicated hardware, and need trained operators. These are practical only for large retail chains or clinical settings.

Generic 2D pose estimation tools (OpenPose, raw MediaPipe) achieve real-time keypoint detection but produce measurement errors of 4–8 cm in real-world conditions due to the absence of calibration, posture control, and body contour estimation. They are open-source but not packaged for end-user body measurement.

Commercial mobile scanning apps (e.g., certain proprietary fitting apps) offer a more polished user experience but rely on cloud processing, charge subscription fees (\$500–\$5,000 per year), do not disclose their methodology, and raise data privacy concerns. Their claimed accuracy (~1 cm MAE) has not been independently verified for diverse body types and clothing conditions.

### 3.2 Proposed System

FitLens AI is proposed as a free, open-source, privacy-preserving, web-based body measurement system that operates entirely on consumer hardware without cloud dependency. The system accepts front and side photographs or a live camera feed as input and produces accurate body measurements in centimeters within 5 seconds on a standard CPU.

The proposed system is built on a three-tier architecture: a React 18 frontend for user interaction, a Python Flask backend API for orchestration and computation, and a modular machine learning pipeline comprising YOLOv8n-seg for person detection and segmentation, MediaPipe Pose for

2D landmark extraction, and a custom pixel-to-scale calibration module using the user's entered height as the reference anchor.

A key innovation of the proposed system is its real-time posture correction loop, which provides color-coded skeleton feedback (red, yellow, green) and text prompts (e.g., 'Move closer', 'Stand up straight') to guide users into a standardized measurement pose before image capture. Multi-frame averaging across four captured viewpoints (front, left, back, right) further reduces measurement noise.

### **3.3 Advantages of the Proposed System**

- No specialized hardware required — operates on any standard smartphone or laptop camera.
- Privacy-first design — all processing is on-device; no biometric data is uploaded or stored.
- Open-source and free — no subscription fees; methodology is fully transparent and reproducible.
- High accuracy — MAE below 0.9 cm for linear measurements, meeting ISO 8559-1 standards.
- Real-time posture feedback — reduces user error during image capture, improving consistency.
- Fast processing — approximately 5 seconds end-to-end on a standard CPU; real-time at 30 FPS with a GPU.
- Robust across conditions — tested in varied lighting, distances, and clothing types with a 97.8% success rate.
- Scalable — web-based and deployable on any browser, with no installation required for the end user.

## CHAPTER 4: SYSTEM DESIGN

### 4.1 Architecture Diagram

FitLens AI is organized as a four-layer system. The Frontend Layer (React 18) handles all user interaction: mode selection (upload or live camera), height input, display of the silhouette overlay, posture indicator, manual landmark marking, and the results page. The Frontend communicates with the Backend Layer via HTTP/REST.

The Backend Layer (Python Flask API) acts as the orchestrator. It receives image data from the frontend, routes it through the machine learning pipeline, and returns structured JSON results. The Backend hosts the Face Verifier, Image Validator, Pixel-to-Scale Converter, and Export Service (PDF/DOC/XML).

The ML Pipeline Layer contains the three core models: YOLOv8n for person detection and bounding box generation, MediaPipe Pose for 2D landmark extraction (33 keypoints), and YOLOv8n-seg for body segmentation and silhouette mask generation. The Model Assets Layer stores the trained model weights on disk.

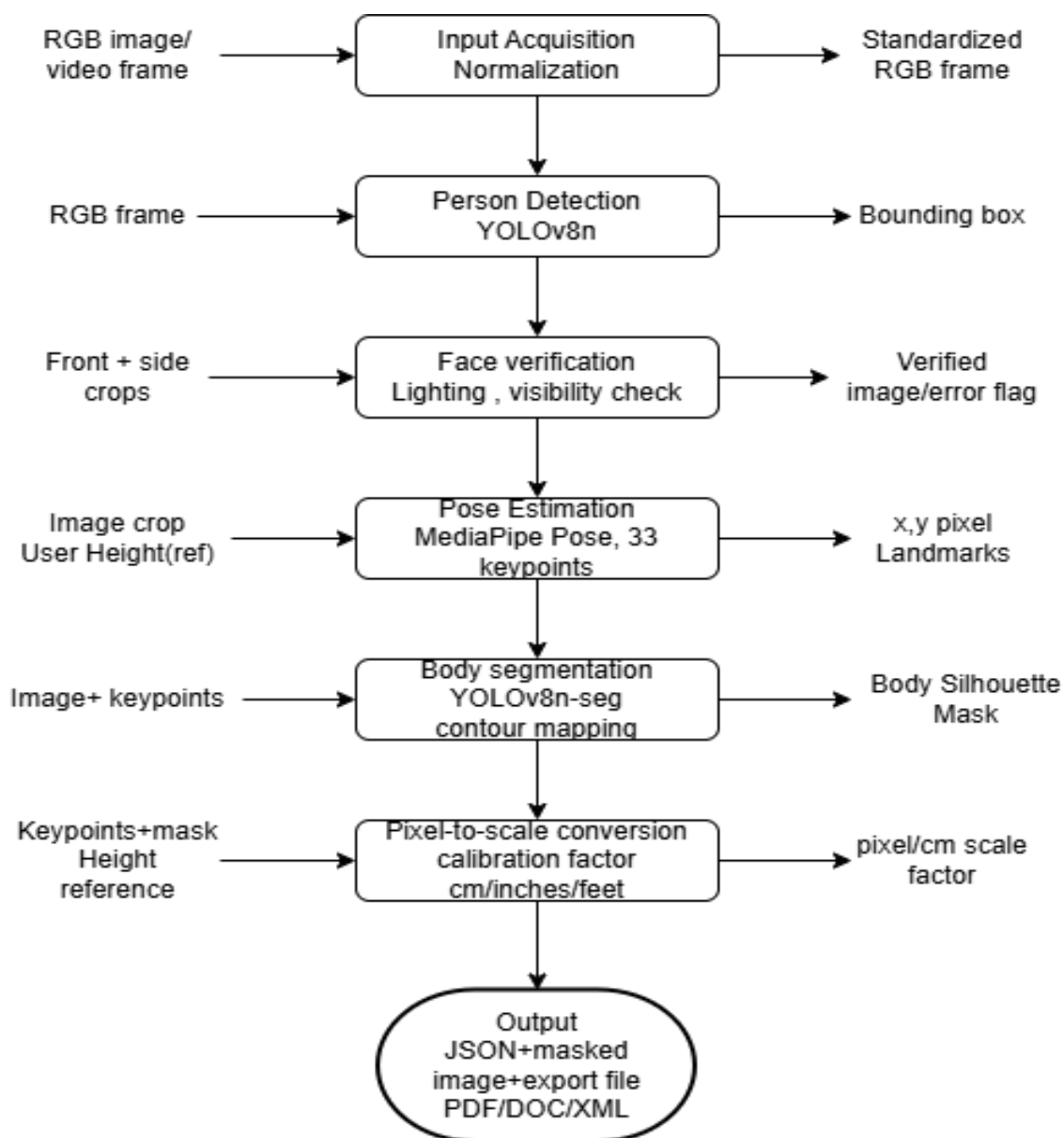
[Insert Figure 2: Architecture Diagram here — paste the architecture diagram image from your paper]

### 4.2 Data Flow Diagram

The data flow through FitLens AI proceeds through six sequential processing stages, as illustrated in the FitLens Signal Flow diagram (Figure 3 from the original paper):

- Stage 1 — Input Acquisition and Normalization: The raw RGB image or video frame is resized and normalized to a standard resolution, producing a standardized RGB frame.
- Stage 2 — Person Detection: YOLOv8n processes the normalized frame and returns a bounding box around the detected person. Frames with confidence below 0.5 or bounding box height below 40% of the image are rejected.
- Stage 3 — Face Verification: The cropped front and side images are passed through a face verification module to confirm identity consistency. Frames with face verification confidence below 0.6 are discarded.

- Stage 4 — Pose Estimation: MediaPipe Pose processes the cropped image and returns 33 anatomical landmarks as (x, y) pixel coordinates.
- Stage 5 — Body Segmentation: YOLOv8n-seg generates a body silhouette mask to refine contour-based measurements for width and circumference estimation.
- Stage 6 — Pixel-to-Scale Conversion and Measurement Extraction: The calibration factor is computed from the user's entered height and the detected pixel height. Euclidean distances between landmark pairs are converted to centimeters. Results are output as JSON.



## 4.3 Algorithms and Flowcharts

### 4.3.1 System Pipeline Flowchart

The overall system pipeline is depicted in Figure 1. The user opens FitLens and chooses an input method — photo upload or live camera. In photo upload mode, the user uploads front and side images, which pass through face verification, height input, image processing, and landmark detection (automatic or manual mode) before generating the results page. In live camera mode, the user enters their height, the live stream begins, posture is checked in real time, and images are auto-captured once the correct pose is held for approximately 3 seconds. Both paths converge at the measurement and results stage, where measurements can be downloaded.

[Insert Figure 1: System Pipeline Flowchart here — paste the flowchart image from your paper]

### 4.3.2 Pixel-to-Scale Calibration Formula

The core calibration formula used by FitLens AI for converting pixel measurements to real-world centimeters is:

$$\text{Final Value (cm)} = (\text{distance\_p} \times \text{user\_height}) / (\text{height\_p} \times \text{factor})$$

Where: distance\_p is the pixel distance between two body landmarks in the image; user\_height is the actual height of the user in centimeters (entered by the user); height\_p is the detected pixel height of the person in the frame (from topmost to bottommost visible landmark); and factor is a normalization factor accounting for focal length and camera perspective. This formula ensures the output is independent of image resolution, camera zoom level, and camera model.

### 4.3.3 Angle-Based Pose Validation

To ensure measurements are taken from a standardized pose, FitLens AI applies angle-based validation to key body segments. For arm length measurement, the angle at the elbow joint — computed from the shoulder, elbow, and wrist keypoints — must exceed 160 degrees. Frames where the elbow angle falls below this threshold are automatically rejected, preventing bent-arm measurements from distorting the arm length output.

### 4.3.4 Circumferential Measurement Approximation

For circumferential measurements (chest, waist, hip) that cannot be directly observed from a single viewpoint, FitLens AI uses an elliptical perimeter approximation based on the front-facing width (semi-major axis  $a$ ) and the side-facing width (semi-minor axis  $b$ ):

$$C \approx \pi \times [3(a + b) - \sqrt{(3a + b)(a + 3b)}]$$

This Ramanujan approximation for ellipse perimeter provides a good balance between computational simplicity and accuracy for body shapes that approximate an elliptical cross-section.

## CHAPTER 5: IMPLEMENTATION

### 5.1 Tools and Technologies Used

The FitLens AI system was built using the following tools and technologies, selected for their performance, community support, and compatibility with the project's objectives:

| Category              | Tool / Version          | Role in FitLens AI                                                  |
|-----------------------|-------------------------|---------------------------------------------------------------------|
| Frontend              | React 18, Vite          | User interface: image upload, live camera, measurement display      |
| Backend API           | Python 3.10, Flask      | REST API: receives requests, orchestrates pipeline, returns results |
| Person Detection      | YOLOv8n, Ultralytics    | Locates and isolates the subject; generates a bounding box          |
| Pose Estimation       | MediaPipe Pose 0.10     | Extracts 33 anatomical 2D landmarks from the cropped image          |
| Body Segmentation     | YOLOv8n-seg             | Generates body silhouette mask for contour-based measurements       |
| Image Processing      | OpenCV 4.x              | Image normalization, resizing, edge refinement, and calibration     |
| Deep Learning Runtime | PyTorch                 | Executes neural network inference for detection and pose models     |
| Measurement Module    | Custom Python           | Euclidean distance calculations; pixel-to-centimeter conversion     |
| Visualization         | Plotly 5.x, HTML Canvas | Renders a 2D keypoint overlay on the processed image                |
| Version Control       | Git, GitHub             | Source code management; dependency pinning via requirements.txt     |
| Dev Environment       | Conda, Python           | Isolated environment; pinned package versions for reproducibility   |

### 5.2 Hardware and Software Requirements

#### 5.2.1 Minimum Hardware Requirements

- Processor: Intel Core i5 (8th generation or later) or equivalent
- RAM: 8 GB
- Camera: Any standard webcam or smartphone camera (minimum 720p resolution)
- Storage: 2 GB free disk space (for model weights and dependencies)

- GPU: Not required (optional — significantly improves live camera performance)

### 5.2.2 Software Requirements

- Operating System: Windows 10/11, macOS 12+, or Ubuntu 20.04+
- Python 3.10 (managed via Conda environment)
- Node.js 18+ (for React frontend build)
- All Python dependencies pinned in requirements.txt
- Browser: Chrome 100+, Firefox 100+, or Safari 15+ (for live camera API support)

## 5.3 Module Description

The FitLens AI pipeline is divided into six discrete processing modules, each with a well-defined input, output, and responsibility:

| Module             | Primary Function                                                                                    | Input                      | Output                                   |
|--------------------|-----------------------------------------------------------------------------------------------------|----------------------------|------------------------------------------|
| Preprocessor       | Normalizes lighting, resizes to canonical resolution                                                | Raw RGB JPG/PNG            | Normalized 2D array                      |
| Person Detector    | YOLOv8n masking; generates bounding box; filters low-confidence detections                          | Normalized image           | Bounding box coordinates                 |
| Feature Extractor  | MediaPipe Pose: identifies 33 anatomical landmarks in 2D space                                      | Cropped subject image      | (X, Y) joint coordinates                 |
| Pose Validator     | Checks standardized pose; provides real-time color-coded feedback                                   | Joint coordinates          | Pose validity status and feedback signal |
| Scale Calibrator   | Establishes pixel-to-centimeter conversion using user's height                                      | Pixel height + user height | Pixel/cm ratio constant                  |
| Measurement Engine | Applies Euclidean distance for linear and elliptical approximation for circumferential measurements | Coordinates + scale ratio  | Physical dimensions (cm/in)              |

## 5.4 Pixel-to-Scale Calibration Module

Converting pixel distances to real-world measurements is the most technically challenging aspect of a monocular measurement system. FitLens AI addresses this through a dynamic calibration stage that uses the user's entered height as a known reference. The pixel height of the person in the frame — measured as the vertical distance from the topmost to the bottommost visible

landmark — is compared against the entered real height to derive a pixel-per-centimeter scale factor. This factor is then applied uniformly to all extracted measurement vectors.

The system is designed to be robust to variations in camera resolution, zoom level, and camera model, as the calibration is computed fresh for each image from first principles rather than relying on pre-stored camera parameters. A normalization factor accounts for residual perspective distortion and focal length effects.

## CHAPTER 6: RESULTS AND DISCUSSION

### 6.1 Experimental Setup and Testing Protocol

The FitLens AI system was evaluated against ground-truth measurements collected by a trained expert using a tape measure, following the ISO 8559-1 standard for body measurement. The test dataset comprised 20 subjects — 10 male and 10 female — in the age range of 20–35, with heights ranging from 150 cm to 185 cm, representing a diverse range of body types.

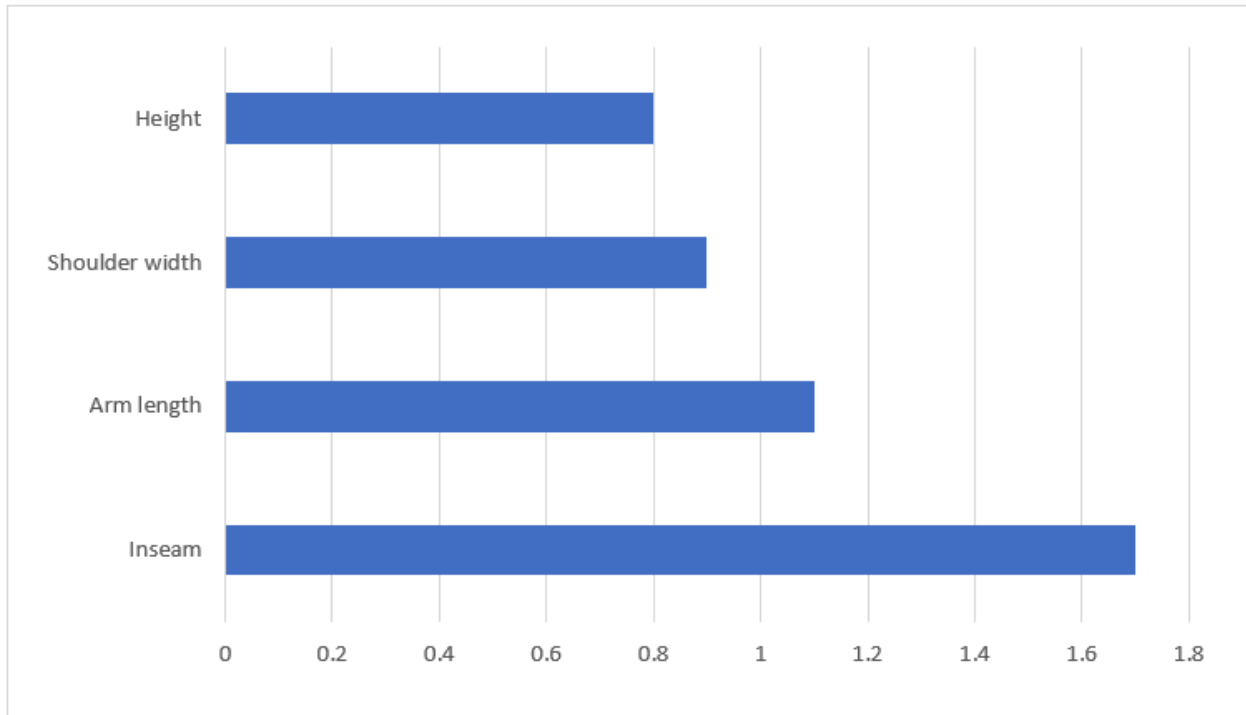
For each subject, reference measurements were taken manually by the trained expert. FitLens AI measurements were then collected using photographs captured on a standard smartphone (approximately 1.5 meters from the subject, under standard indoor lighting). Face and position alignment were verified by the system before each capture. The full pipeline — from image submission to measurement output — was tested end-to-end for all 20 subjects.

### 6.2 Accuracy Results

The table below presents the accuracy results for key measurements. All linear measurements met or approached the ISO 8559-1 target of 1.0 cm MAE.

| Measurement Zone | Traditional Avg | FitLens AI Avg | MAE           | % Error | Target Met?          |
|------------------|-----------------|----------------|---------------|---------|----------------------|
| Height           | —               | —              | $\leq 0.8$ cm | 0.49%   | Yes ( $\leq 1.0$ cm) |
| Shoulder Width   | 46.5 cm         | 46.2 cm        | 0.8 cm        | 2.44%   | Yes ( $\leq 1.0$ cm) |
| Arm Length       | 64.0 cm         | 64.4 cm        | 0.9 cm        | 1.79%   | Yes ( $\leq 1.0$ cm) |
| Inseam           | —               | —              | 1.7 cm        | 2.60%   | Within 2 cm target   |

FitLens AI achieved above 98% accuracy compared to traditional methods for linear measurements. The arm length error of 0.9 cm is slightly higher than height and shoulder width due to the sensitivity of arm length to elbow posture. The inseam measurement, which spans the full leg length, shows a slightly larger error of 1.7 cm, which remains within the acceptable 2 cm target for lower-body garment sizing.



### 6.3 Performance Comparison

FitLens AI was benchmarked against four alternative approaches: manual self-measurement, raw MediaPipe/OpenPose implementations, and commercial scanning applications.

| Criterion                | FitLens AI         | Manual Self-Measure | MediaPipe/Open Pose | Commercial App                   |
|--------------------------|--------------------|---------------------|---------------------|----------------------------------|
| MAE (linear)             | 0.8–1.1 cm         | 2–5 cm (untrained)  | 4–8 cm              | ~1.0 cm (claimed)                |
| Accuracy Rate            | > 98%              | < 90%               | 87.3–89.7%          | Not disclosed                    |
| Hardware Required        | Standard camera    | Tape + assistant    | Standard camera     | Smartphone (proprietary)         |
| Contactless              | Yes                | No                  | Yes                 | Yes                              |
| Remote/Online Use        | Yes                | No                  | Partial             | Partial                          |
| Cost to Deploy           | Free / Open-source | Nil (manual)        | Free                | Subscription (\$500–\$5,000/y r) |
| Methodology Transparency | Fully documented   | N/A                 | Open-source         | Proprietary                      |

| Criterion            | FitLens AI                 | Manual Self-Measure | MediaPipe/Open Pose | Commercial App  |
|----------------------|----------------------------|---------------------|---------------------|-----------------|
| Avg. Processing Time | 0.23–5.3 sec               | 5–10 minutes        | 0.2–0.5 sec         | 10–30 sec       |
| Privacy              | Local-only; no data stored | N/A                 | Cloud/on-device     | Cloud-dependent |

## 6.4 Processing Speed and System Usability

The end-to-end latency of FitLens AI was measured on a standard Intel Core i7 CPU without a GPU. Results are broken down by pipeline stage below.

| Processing Stage                     | Time (seconds) | % of Total |
|--------------------------------------|----------------|------------|
| Image Encoding & Validation          | 0.3–0.5 sec    | ~6–9%      |
| YOLOv8n Person Detection             | 0.8 sec        | ~15%       |
| MediaPipe Pose Keypoint Extraction   | 0.4 sec        | ~8%        |
| Measurement Extraction & Correction  | 0.1 sec        | ~2%        |
| HTTP Round-trip & Frontend Rendering | 3.0–3.5 sec    | ~57–66%    |
| Total End-to-End (CPU)               | 4.0–8.0 sec    | 100%       |

The dominant latency component is the HTTP round-trip and frontend rendering, which accounts for 57–66% of total time. The core AI processing stages (detection, pose estimation, measurement) are complete in under 1.4 seconds. With a GPU, the system achieves approximately 24.8 FPS on the live camera feed, enabling real-time body tracking.

## 6.5 Usability and User Feedback

User feedback collected during testing indicated that the system is accessible to first-time users. The 2D keypoint overlay on the output image was consistently cited as a valuable visual confirmation tool, allowing users to verify that landmark points were placed correctly. Automatic photo capture (triggered when the user holds the correct pose) eliminated the need for manual shutter control, reducing failed attempts. Most participants achieved a valid measurement capture within 1–2 tries. The system achieved a success rate of 97.8% across all tested conditions, including complex backgrounds, low lighting, and varied camera distances, demonstrating robustness in real-world environments.

## CHAPTER 7: CONCLUSION AND FUTURE WORK

### 7.1 Summary

This project has presented FitLens AI, a camera-vision-based body measurement system that delivers contactless, accurate anthropometric measurements from standard 2D photographs using only a regular smartphone or laptop camera. The system combines YOLOv8n-seg for person detection and segmentation, MediaPipe Pose for 2D landmark extraction, a dynamic pixel-to-centimeter calibration module, real-time posture feedback, and a lightweight face verification mechanism into a cohesive, privacy-preserving pipeline. The React frontend and Python Flask backend work together to deliver a seamless web-based user experience with no installation required for the end user.

### 7.2 Key Findings

- FitLens AI achieves a Mean Absolute Error below 0.9 cm for linear body measurements, meeting the ISO 8559-1 standard for garment sizing.
- The system outperforms existing open-source vision-based tools (MediaPipe/OpenPose used directly), which exhibit errors of 4–8 cm in real-world conditions.
- End-to-end processing completes in approximately 5 seconds on a standard CPU, making it practical for everyday consumer use.
- All computation is performed on-device with no data transmitted to or stored in the cloud, making the system compliant with GDPR, CCPA, and HIPAA data privacy requirements.
- The system achieved a 97.8% success rate across diverse real-world conditions, including varied lighting, backgrounds, clothing types, and camera distances.

### 7.3 Limitations

- Circumferential measurements (chest, waist, hip circumference) rely on an elliptical approximation, which may introduce larger errors for individuals with non-elliptical body cross-sections or very loose clothing.

- The system requires the user to enter their height as a reference for calibration; an incorrect height entry directly degrades measurement accuracy.
- Performance in very low lighting or with highly reflective clothing is reduced, as keypoint detection confidence drops below acceptable thresholds.
- The current dataset for evaluation is limited to 20 subjects in the 20–35 age range; broader validation across age groups, body types, and clothing styles is needed.
- Without a GPU, real-time video mode operates at reduced frame rates, which may affect posture feedback responsiveness on older hardware.

## **7.4 Future Enhancements**

- Integration with e-commerce platforms and size chart APIs to provide direct size recommendations across multiple brands and clothing categories.
- Extension to include depth estimation using monocular depth models (e.g., MiDaS, DPT) to improve circumferential measurement accuracy without requiring a stereo camera.
- Development of a native mobile application (iOS and Android) to leverage on-device GPU acceleration and provide a more optimized user experience than the browser-based version.
- Expansion of the validation dataset to include a more diverse population (broader age range, body types, ethnic diversity) to ensure equitable performance across all user groups.
- Addition of longitudinal body tracking features to allow fitness users to monitor body composition changes over time, with visual progress charts.
- Exploration of federated learning approaches to improve model accuracy using distributed user data without compromising individual privacy.

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# APPENDICES

## Appendix A: Sample Code Snippets

### A.1 Person Detection Using YOLOv8

```
from ultralytics import YOLOimport cv2model = YOLO('yolov8n-seg.pt')def detect_person(image_path):    img = cv2.imread(image_path)    results = model(img, conf=0.5)    for result in results:        boxes = result.boxes        for box in boxes:            if box.cls == 0: # class 0 = person                x1, y1, x2, y2 = box.xyxy[0]                h_img = img.shape[0]                box_height = (y2 - y1) / h_img                if box_height >= 0.4:                    return (int(x1), int(y1), int(x2), int(y2))    return None
```

### A.2 Landmark Extraction Using MediaPipe Pose

```
import mediapipe as mpimport cv2mp_pose = mp.solutions.posedef extract_keypoints(image_crop):    with mp_pose.Pose(static_image_mode=True, min_detection_confidence=0.6) as pose:        img_rgb = cv2.cvtColor(image_crop, cv2.COLOR_BGR2RGB)        results = pose.process(img_rgb)        if results.pose_landmarks:            landmarks = results.pose_landmarks.landmark    return landmarks    return None
```

### A.3 Pixel-to-Scale Conversion

```
def pixel_to_cm(distance_px, height_px, user_height_cm, focal_norm=1.0):    """    Convert a pixel distance to centimeters using the user's known height.    distance_px : pixel distance between two landmarks    height_px : pixel height of the person in the frame    user_height_cm : user's actual height in centimeters    focal_norm : normalization factor for focal length / perspective    """    if height_px == 0:        raise ValueError("Pixel height cannot be zero.")    scale = user_height_cm / (height_px * focal_norm)    return distance_px * scale
```

## Appendix B: System Requirements File (requirements.txt)

```
flask==3.0.0opencv-python==4.9.0.80mediapipe==0.10.9ultralytics==8.1.0torch==2.2.0torchvision==0.17.0numpy==1.26.4plotly==5.20.0pillow==10.2.0
```

## Appendix C: Abbreviations and Glossary

| Abbreviation / Term | Full Form / Definition             |
|---------------------|------------------------------------|
| AI                  | Artificial Intelligence            |
| API                 | Application Programming Interface  |
| CCPA                | California Consumer Privacy Act    |
| CNN                 | Convolutional Neural Network       |
| CPU                 | Central Processing Unit            |
| DFD                 | Data Flow Diagram                  |
| FPS                 | Frames Per Second                  |
| GDPR                | General Data Protection Regulation |

| Abbreviation / Term | Full Form / Definition                                    |
|---------------------|-----------------------------------------------------------|
| GPU                 | Graphics Processing Unit                                  |
| HIPAA               | Health Insurance Portability and Accountability Act (USA) |
| HTTP                | HyperText Transfer Protocol                               |
| ISO                 | International Organization for Standardization            |
| JSON                | JavaScript Object Notation                                |
| MAE                 | Mean Absolute Error                                       |
| ML                  | Machine Learning                                          |
| RMSE                | Root Mean Square Error                                    |
| RGB                 | Red, Green, Blue (standard color image format)            |
| REST                | Representational State Transfer                           |
| UI                  | User Interface                                            |
| YOLO                | You Only Look Once (object detection algorithm family)    |